PRACTICE EXAM 1 · EXAM STRATEGY - ALL DOMAINS

Practice exam 1

Twenty questions, weighted to the real section proportions. Forty minutes if you are pacing properly.

TYPE Practice exam QUESTIONS 20 TOPIC All sections

admin@sthithapragna.com

How to get value from this

- **Pause on each question slide and commit to an answer** before continuing. Reading ahead teaches you nothing.
- Write your answers down. You want to know what you chose, not what you would have chosen.
- **A wrong answer is the most useful thing on the page.** Read the reasoning for every option, not only the right one.
- If you get one right for the wrong reason, count it as wrong. That is the one that will catch you on the day.

> ON THE EXAM
>
> Score interpretation comes at the end. Do not look at it first, and do not stop partway to check how you are doing.

> A team packages shared transformation logic as a Python wheel and deploys it with a bundle. Several notebook tasks import it. After a change to the wheel, some tasks pick up the new version and others continue using the old one within the same job run.
>
> The tasks run on a shared all-purpose cluster.

What is the most likely cause?

A. The wheel was not rebuilt, so both versions are identical.

B. The library is installed at cluster scope and already loaded into running Python processes.

C. Bundles cannot deploy wheels to all-purpose compute.

D. Notebook tasks always pin the wheel version at first import.

◆ PAUSE AND DECIDE

What is the most likely cause?

A The wheel was not rebuilt, so both versions are identical.

Then every task would behave the same. The scenario says they differ.

B The library is installed at cluster scope and already loaded into running Python processes.

A cluster-scoped library loaded into a live process is not swapped underneath it. Job compute avoids this entirely.

C Bundles cannot deploy wheels to all-purpose compute.

They can. The compute type is the risk, not a restriction.

D Notebook tasks always pin the wheel version at first import.

There is no such pinning behaviour.

ON THE EXAM

The shared all-purpose cluster is the clue. Fresh job compute per run gives each task a clean environment.

> A transformation applies a business rule to 3 billion rows. It is currently implemented as a Python user-defined function and takes four hours. The rule is a series of conditional string comparisons producing a category label.
>
> The team asks how to make it faster.

What is the first thing to try?

A Rewrite it as a pandas UDF, which vectorises the operation.

B Express the rule in native SQL, since conditional logic on strings needs no UDF at all.

C Increase the number of shuffle partitions so the UDF runs on more tasks.

D Register the UDF in Unity Catalog so it can be optimised.

◆ PAUSE AND DECIDE

What is the first thing to try?

A Rewrite it as a pandas UDF, which vectorises the operation.

A real improvement over a row-at-a-time Python UDF, and still slower than not leaving the engine at all.

B Express the rule in native SQL, since conditional logic on strings needs no UDF at all.

Conditional string logic is expressible in SQL, which keeps execution in the engine and avoids serialisation entirely.

C Increase the number of shuffle partitions so the UDF runs on more tasks.

Parallelises the serialisation overhead rather than removing it.

D Register the UDF in Unity Catalog so it can be optimised.

Registration governs and shares a function. It does not change how it executes.

ON THE EXAM

A is the trap because it is genuinely better. The question asks for the first thing to try, and the first thing is to stop leaving the engine.

> A pipeline computes a daily aggregate of orders by region. The source is a Delta table that receives late-arriving corrections to orders from up to two weeks earlier. The business needs the aggregate to always be correct.
>
> An engineer proposes a streaming table for cost reasons.

What should they use, and why?

A. A streaming table, which reprocesses corrections automatically.

B. A materialized view, which always returns the correct result and recomputes when incremental processing cannot.

C. A streaming table with a two-week watermark.

D. A materialized view is unnecessary; add a scheduled full refresh to the streaming table.

◆ PAUSE AND DECIDE

What should they use, and why?

A A streaming table, which reprocesses corrections automatically.

A streaming table processes each row once and does not revisit it. Corrections to old rows are not reprocessed.

B A materialized view, which always returns the correct result and recomputes when incremental processing cannot.

Correctness is the stated requirement, and a materialized view never trades it for speed.

C A streaming table with a two-week watermark.

A watermark bounds state in a streaming aggregation. It does not make a streaming table revisit processed rows.

D A materialized view is unnecessary; add a scheduled full refresh to the streaming table.

A full refresh recomputes everything nightly, which costs more than the materialized view being avoided.

ON THE EXAM

Always correct is the requirement sentence. A streaming table's process-once guarantee is exactly what the corrections violate.

A supplier dimension is maintained with AUTO CDC as Type 2. A supplier is deleted in the source system, and the delete condition is configured.

An analyst reports that the supplier has vanished from the dimension entirely, and asks whether the history was lost.

What actually happened?

A The history was deleted along with the record, as Type 2 deletes remove all versions.

B The current version was closed with an end value; the history remains. The analyst is filtering on a null end date.

C AUTO CDC does not support deletes on Type 2 targets.

D The supplier was moved to the rescued data column.

◆ PAUSE AND DECIDE

What actually happened?

A The history was deleted along with the record, as Type 2 deletes remove all versions.

Type 2 preserves history through deletion. That is much of the point of it.

B The current version was closed with an end value; the history remains. The analyst is filtering on a null end date.

A deleted record has its final version closed rather than removed, so it disappears from a current-state filter.

C AUTO CDC does not support deletes on Type 2 targets.

It supports them through the delete condition, which the scenario says is configured.

D The supplier was moved to the rescued data column.

Rescued data captures records that did not parse. This one parsed and was deleted.

ON THE EXAM

Vanished from the dimension and vanished from the analyst's query are different claims. The second is what happened.

A job must process a different set of country files each night, determined by which countries delivered data. The list changes daily. A first task queries the landing zone and produces the list.

An engineer configures a For each task fed by that list, and the job fails immediately with nothing having run.

What is missing?

A. A nested task inside the For each task. It has nothing to run.

B. A task value must be set explicitly; For each cannot read a task's output.

C. The For each task requires a static array; dynamic lists are unsupported.

D. An If/else task must precede the For each to check the list is non-empty.

◆ PAUSE AND DECIDE

What is missing?

A A nested task inside the For each task. It has nothing to run.

A For each task requires both itself and a nested task. Neither does anything alone.

B A task value must be set explicitly; For each cannot read a task's output.

Dynamically generated input arrays are supported and are the powerful case.

C The For each task requires a static array; dynamic lists are unsupported.

The opposite. Generating the array dynamically is the documented pattern.

D An If/else task must precede the For each to check the list is non-empty.

Sensible defensive design, and not what causes an immediate failure.

ON THE EXAM

The structural trap from module 9. A For each task is half a construction, and the other half is a nested task.

> An Auto Loader stream ingests CSV files with an explicit schema supplied at creation. A source adds two new columns. The stream continues without error, and the new columns never appear downstream.
>
> The team expected the stream to fail so they would notice.

Why did nothing happen?

A Supplying a schema sets the evolution mode default to none, which ignores new columns without failing.

B CSV does not support schema evolution.

C The new columns were written to the rescued data column, which nobody queried.

D Schema evolution only triggers when a column's type changes, not when one is added.

◆ PAUSE AND DECIDE

Why did nothing happen?

A Supplying a schema sets the evolution mode default to none, which ignores new columns without failing.

The two-defaults rule. With a schema supplied the default is none, which ignores new columns and rescues nothing.

B CSV does not support schema evolution.

It does. CSV is one of the formats where evolution matters most.

C The new columns were written to the rescued data column, which nobody queried.

None mode does not rescue unless a rescued data column is explicitly requested.

D Schema evolution only triggers when a column's type changes, not when one is added.

Adding a column is the canonical evolution case.

ON THE EXAM

The team's expectation was reasonable and describes addNewColumns, which is the default only when no schema is supplied.

> A query must return, for each product, the most recent price change and the price immediately before it. An engineer writes a window function with a partition by product and an order by change date, then filters on the row number in a WHERE clause.
>
> The query fails to resolve the column.

What is the cleanest fix on Databricks?

A Move the filter to HAVING.

B Use QUALIFY, which filters window results in the same query block.

C Replace row_number with dense_rank, which is resolvable in WHERE.

D Add the row number to the GROUP BY.

◆ PAUSE AND DECIDE

What is the cleanest fix on Databricks?

A Move the filter to HAVING.

Window functions are evaluated after HAVING too.

B Use QUALIFY, which filters window results in the same query block.

Exactly what QUALIFY exists for, and it is Databricks-specific, which is why the question says on Databricks.

C Replace row_number with dense_rank, which is resolvable in WHERE.

No ranking function is resolvable in WHERE. The function is not the problem.

D Add the row number to the GROUP BY.

Collapses the rows being selected.

ON THE EXAM

A CTE would also work and is not offered. The phrase 'on Databricks' is pointing at the platform-specific answer.

> A full outer join between two large tables is slow. The Spark UI shows median task duration of six seconds and maximum of forty minutes, with shuffle read similarly lopsided.
>
> An engineer says adaptive query execution will split the skewed partition automatically once statistics refresh.

What is the correct assessment?

A Correct. AQE splits skewed partitions in shuffle-based joins once statistics are current.

B AQE does not skew-optimise a FULL OUTER JOIN on either side. A manual approach such as salting is required.

C AQE handles this, but only if the join is converted to a broadcast join first.

D The maximum-to-median ratio indicates memory pressure, not skew.

◆ PAUSE AND DECIDE

What is the correct assessment?

A — Correct. AQE splits skewed partitions in shuffle-based joins once statistics are current.

Nearly right, and the join type is the exception. AQE also uses runtime statistics, not collected ones.

B — AQE does not skew-optimise a FULL OUTER JOIN on either side. A manual approach such as salting is required.

A documented limit. Skew handling applies to shuffle joins, and full outer is excluded on both sides.

C — AQE handles this, but only if the join is converted to a broadcast join first.

Broadcast joins are never skew-optimised, and both tables are large.

D — The maximum-to-median ratio indicates memory pressure, not skew.

Uniformly slow tasks indicate resources. A few far slower than the median is skew.

ON THE EXAM

A is wrong twice: the join type is excluded, and AQE works from runtime statistics precisely so it does not wait for collected ones.

> A healthcare provider shares a patient table with a research partner using WITH HISTORY for performance. Records are routinely deleted under right-to-erasure requests. Vacuum runs on the default retention.
>
> A privacy review asks what the recipient can currently read.

What is the accurate answer?

A Only the current table contents, since deletes are applied immediately.

B The current contents plus, via the Delta log, commit history and deleted data not yet vacuumed.

C Nothing beyond current contents; recipients cannot access the Delta log.

D Everything ever written, since WITH HISTORY disables vacuum.

◆ PAUSE AND DECIDE

What is the accurate answer?

A Only the current table contents, since deletes are applied immediately.

True of a query against the table, and not of direct file access to its directory.

B The current contents plus, via the Delta log, commit history and deleted data not yet vacuumed.

WITH HISTORY grants cloud tokens scoped to the table's root directory, which includes the log.

C Nothing beyond current contents; recipients cannot access the Delta log.

Cloud token access is precisely access to the directory, log included.

D Everything ever written, since WITH HISTORY disables vacuum.

It does not disable vacuum. The window is bounded by retention.

ON THE EXAM

The vacuum window bounds the exposure, which is why D overstates it and A understates it. The truthful answer sits between them.

A cost report built from the billable usage system table shows one job costing far more than the team expects. The report joins usage to the jobs table on job identifier and sums the usage quantity.

That job's definition has been edited eleven times this year. Other jobs, rarely edited, look correct.

What is wrong?

A. The job genuinely costs more because each edit redeploys it.

B. The jobs table is SCD Type 2, so the join multiplies usage by the number of job versions.

C. Usage records for edited jobs are duplicated by the billing system.

D. The report must filter to record type ORIGINAL to avoid double counting.

◆ PAUSE AND DECIDE

What is wrong?

A The job genuinely costs more because each edit redeploys it.

Editing a definition does not run it.

B The jobs table is SCD Type 2, so the join multiplies usage by the number of job versions.

Every edit emits a new row, and joining without restricting to one version multiplies the usage.

C Usage records for edited jobs are duplicated by the billing system.

Usage is recorded per run. The duplication is in the join.

D The report must filter to record type ORIGINAL to avoid double counting.

That filter causes overcounting by discarding retractions, and is a different trap entirely.

ON THE EXAM

The correlation with edit count is the tell, and D is a real trap from the same table applied to the wrong symptom.

A job runs a SQL alert task checking that a loaded table's null rate is below a threshold, then a publication task depending on it. On a night when the null rate breached the threshold, the alert notified the team and publication proceeded.

The dependency uses the default run-if condition.

How should the job be changed?

A Change the run-if condition to None failed.

B Branch on the alert task's alert state with an If/else task, since task status reports whether the alert evaluated, not what it found.

C Configure the alert to fail the task when triggered.

D Move the alert to a schedule so it runs before the job.

◆ PAUSE AND DECIDE

How should the job be changed?

A Change the run-if condition to None failed.
Loosens the condition. The alert task succeeded either way.

B Branch on the alert task's alert state with an If/else task, since task status reports whether the alert evaluated, not what it found.
Task status is Succeeded whenever evaluation completed. The verdict is a value, so it needs If/else.

C Configure the alert to fail the task when triggered.
No such setting. Failed means an evaluation error.

D Move the alert to a schedule so it runs before the job.
Decouples the check from the load, so the two can disagree about which data was checked.

ON THE EXAM

Module 9's distinction: run-if branches on outcome, If/else branches on a value. The alert's verdict is a value.

> An organisation has 400 external Delta tables and no automatic maintenance. They run a nightly OPTIMIZE and VACUUM job costing a significant amount. A proposal suggests converting the tables to Unity Catalog managed and retiring the job.
>
> Historic time travel of 60 days must be preserved on all tables.

What must they do, and in what order?

A. Convert with SET MANAGED, then set the deleted file retention property to cover 60 days.

B. Set the deleted file retention property to cover 60 days first, then convert and enable predictive optimization.

C. Convert using CREATE TABLE AS SELECT, which preserves history.

D. Keep the tables external; predictive optimization works on external tables too.

◆ PAUSE AND DECIDE

What must they do, and in what order?

A Convert with SET MANAGED, then set the deleted file retention property to cover 60 days.

Right operation, wrong order. Automatic vacuum may run on the 7-day default before the property is set.

B Set the deleted file retention property to cover 60 days first, then convert and enable predictive optimization.

The documented sequence: set retention before enabling, so automatic vacuum respects the longer window.

C Convert using CREATE TABLE AS SELECT, which preserves history.

CTAS creates a new table with no history at all, which is the opposite of the requirement.

D Keep the tables external; predictive optimization works on external tables too.

It runs only on Unity Catalog managed tables.

ON THE EXAM

A and B differ only in order, and the order is the answer. The docs say set retention before enabling.

> A privacy team requires that records deleted under erasure requests be unrecoverable within 30 days. Deletion vectors are enabled on all tables. The engineering team runs DELETE statements and confirms the rows no longer appear in queries.
>
> Predictive optimization is enabled on all tables.

What still needs to happen?

A Nothing further; predictive optimization runs VACUUM, which removes the files.

B REORG TABLE with the purge option, so the files are rewritten without the soft-deleted rows, then vacuum removes the old files.

C Disable deletion vectors and re-run the deletes.

D Set the deleted file retention property to 30 days.

◆ PAUSE AND DECIDE

What still needs to happen?

A Nothing further; predictive optimization runs VACUUM, which removes the files.

Vacuum removes unreferenced files. Files containing soft-deleted rows are still referenced.

B REORG TABLE with the purge option, so the files are rewritten without the soft-deleted rows, then vacuum removes the old files.

With deletion vectors the row stays in the file until a purge rewrites it. The GDPR page requires this step.

C Disable deletion vectors and re-run the deletes.

Purge exists so you keep the write performance and still erase when required.

D Set the deleted file retention property to 30 days.

Lengthening retention keeps files longer, which is the opposite of what erasure needs.

ON THE EXAM

A is the sophisticated wrong answer: vacuum is real and automatic, and it only removes files nothing references.

> A 3 TB unpartitioned table is converted to liquid clustering with keys on region and event date. OPTIMIZE is run and completes. Queries filtering on region show little improvement, though data written after the change clusters well.
>
> The runtime is DBR 16.2.

What should they run?

A OPTIMIZE again, since the first run may have timed out.

B OPTIMIZE FULL, which forces reclustering of all existing data files.

C ZORDER BY region alongside the clustering keys.

D VACUUM, to remove the unclustered files.

◆ PAUSE AND DECIDE

What should they run?

A OPTIMIZE again, since the first run may have timed out.

It completed, and plain OPTIMIZE clusters incrementally however many times you run it.

B OPTIMIZE FULL, which forces reclustering of all existing data files.

Incremental clustering leaves existing files alone. OPTIMIZE FULL is required, and DBR 16.2 supports it.

C ZORDER BY region alongside the clustering keys.

Clustering and Z-order are incompatible, and Z-ordered files are ignored by predictive optimization.

D VACUUM, to remove the unclustered files.

Those files hold current data. Vacuum removes only unreferenced files.

ON THE EXAM

The runtime is in the question for a reason. OPTIMIZE FULL needs DBR 16.0, and 16.2 clears it.

A production job is owned by a platform engineer with broad access across the workspace. To let an on-call rota restart it overnight, the team grants the rota group CAN MANAGE RUN.

A security review asks what the on-call engineers can effectively do.

What is the correct assessment?

A They can start and cancel the job, executing it with their own permissions.

B They can execute the job with the owner's permissions, which are broader than their own.

C They can start the job but it will fail, since they lack access to the underlying data.

D CAN MANAGE RUN grants no data access, so the review has no finding.

◆ PAUSE AND DECIDE

What is the correct assessment?

A They can start and cancel the job, executing it with their own permissions.

True for interactive execution. A job run assumes the owner's permissions.

B They can execute the job with the owner's permissions, which are broader than their own.

A job triggered through Run Now assumes the permissions of the job owner, not the triggering user.

C They can start the job but it will fail, since they lack access to the underlying data.

It will not fail, because it runs as the owner.

D CAN MANAGE RUN grants no data access, so the review has no finding.

Starting a run executes work that reads and writes data.

ON THE EXAM

The fix is service principal ownership scoped to what the job needs, which is why bundles expect one in production mode.

A masking policy is applied to columns tagged as personal data across a catalog. A declarative pipeline whose run-as identity is a service principal reads two of those tables and writes a gold streaming table.

Analysts with full access find the personal data masked in gold, and no grant restores it.

What is the cause and the fix?

A The gold table inherited the policy through lineage. Remove the policy from gold.

B The refresh evaluated policies as the pipeline's run-as identity, writing masked values permanently. Exempt that identity from the policies.

C Streaming tables cannot carry unmasked personal data. Use a materialized view.

D Analysts need EXECUTE on the masking function.

◆ PAUSE AND DECIDE

What is the cause and the fix?

A The gold table inherited the policy through lineage. Remove the policy from gold.

Policies do not propagate through lineage. The masked values are in the data.

B The refresh evaluated policies as the pipeline's run-as identity, writing masked values permanently. Exempt that identity from the policies.

Policies are evaluated using the pipeline owner or run-as identity, and masked values are written in.

C Streaming tables cannot carry unmasked personal data. Use a materialized view.

Invented, and a materialized view would hit the same problem.

D Analysts need EXECUTE on the masking function.

There is nothing to unmask. The original values were never written.

ON THE EXAM

No grant restores it is the tell. Read-time masking is reversible by permission; write-time masking is data loss.

> A data steward is granted MANAGE on a catalog so they can administer permissions across it. They can grant and revoke on every table inside, and transfer ownership.
>
> They report they cannot query any of the tables, and ask whether the grant is incomplete.

What should they be told?

A The grant is incomplete; MANAGE on a catalog should confer SELECT on its tables.

B MANAGE confers administration, not data access. SELECT plus USE CATALOG and USE SCHEMA are still needed.

C They must be the catalog owner rather than a MANAGE holder to query tables.

D MANAGE on a catalog requires USE CATALOG, which they are missing.

◆ PAUSE AND DECIDE

What should they be told?

A The grant is incomplete; MANAGE on a catalog should confer SELECT on its tables.

The separation is deliberate. Administering is not accessing.

B MANAGE confers administration, not data access. SELECT plus USE CATALOG and USE SCHEMA are still needed.

MANAGE holders do not automatically have data privileges, and data access still needs the usage privileges.

C They must be the catalog owner rather than a MANAGE holder to query tables.

An owner has capabilities implied, and MANAGE holders can simply grant themselves what they need.

D MANAGE on a catalog requires USE CATALOG, which they are missing.

MANAGE on a catalog requires no usage privileges. That is the reduced requirement.

ON THE EXAM

D is the near miss: it names the right privilege for the wrong reason. Usage is needed for the SELECT, not for the MANAGE.

> An overnight job on job compute failed. The next morning an engineer finds the job's error message mentions a task failure during a shuffle, opens the driver logs, and sees only routine progress messages.
>
> No log delivery is configured.

What should they do?

A The driver logs are truncated. Increase the log retention setting.

B Worker-side failures are in the executor logs, which are unavailable because the cluster terminated. Configure log delivery for future runs.

C Check the cluster event log, which records task failures.

D Re-run the job with a larger driver, since driver logs are incomplete.

◆ PAUSE AND DECIDE

What should they do?

A The driver logs are truncated. Increase the log retention setting.

They are not truncated. The error is worker-side and was never in them.

B Worker-side failures are in the executor logs, which are unavailable because the cluster terminated. Configure log delivery for future runs.

Both halves: a task failure is worker-side, and job compute terminates taking the Spark UI with it.

C Check the cluster event log, which records task failures.

It records lifecycle events, not task failures.

D Re-run the job with a larger driver, since driver logs are incomplete.

Changes the compute to address a logging misunderstanding.

ON THE EXAM

Two facts in one answer: the right log, and the fact that it is already gone. The remedy is preventive.

> A deployment to a bundle's production target fails validation, reporting that a pipeline is marked as a development pipeline. The engineer confirms the pipeline definition sets development to true, which they added while testing.
>
> They are on the correct Git branch.

What is happening?

A Production mode validates that all deployed pipelines are marked development false. Remove the setting.

B Deploy with the force flag, which overrides production mode validations.

C Switch the target to development mode to deploy successfully.

D Production mode requires a service principal, which is the actual failure.

◆ PAUSE AND DECIDE

What is happening?

A Production mode validates that all deployed pipelines are marked development false. Remove the setting.

Exactly the stated validation, and it is doing its job: a development pipeline cannot reach production by accident.

B Deploy with the force flag, which overrides production mode validations.

Force overrides the Git branch validation specifically, not this one.

C Switch the target to development mode to deploy successfully.

Would prefix every resource and mark pipelines as development, changing production rather than deploying to it.

D Production mode requires a service principal, which is the actual failure.

It recommends one. Without it you get extra validations, not this error.

ON THE EXAM

B is the near miss. The force flag exists, and it overrides the branch check rather than every production validation.

A gold star schema is built with primary keys on dimensions and foreign keys from the fact table. A data quality report finds fact rows whose customer key does not exist in the customer dimension.

The team is surprised, since the foreign key constraint is in place and the load succeeded.

What explains it?

A The constraint was created but not validated. Run a validation command.

B Primary and foreign key constraints are informational and not enforced at write time.

C Foreign keys are enforced only when both tables are in the same schema.

D The dimension is Type 2, so the surrogate keys change and break the reference.

◆ PAUSE AND DECIDE

What explains it?

A The constraint was created but not validated. Run a validation command.

There is no validation step that makes a foreign key enforced.

B Primary and foreign key constraints are informational and not enforced at write time.

They document relationships and inform the optimizer. They never reject a row.

C Foreign keys are enforced only when both tables are in the same schema.

They are not enforced in any arrangement.

D The dimension is Type 2, so the surrogate keys change and break the reference.

Type 2 adds versions with new surrogate keys. Existing keys remain valid.

ON THE EXAM

Contrast with NOT NULL and CHECK on the same table, which are enforced. Enforcement is per constraint type.

How to read your result

- **17 or more.** Comfortable. Work exams 2 and 3, and revisit anything you got right by guessing.
- **13 to 16.** On track. Identify which sections your misses cluster in and revisit those modules.
- **Below 13.** More study needed before booking. Start with Section 1, which is 22% of the paper.
- No pass mark is published, so treat these bands as guidance rather than prediction.

> ON THE EXAM
>
> The more useful number is how many you got right for the right reason. Go back and check the reasoning on every one you guessed.